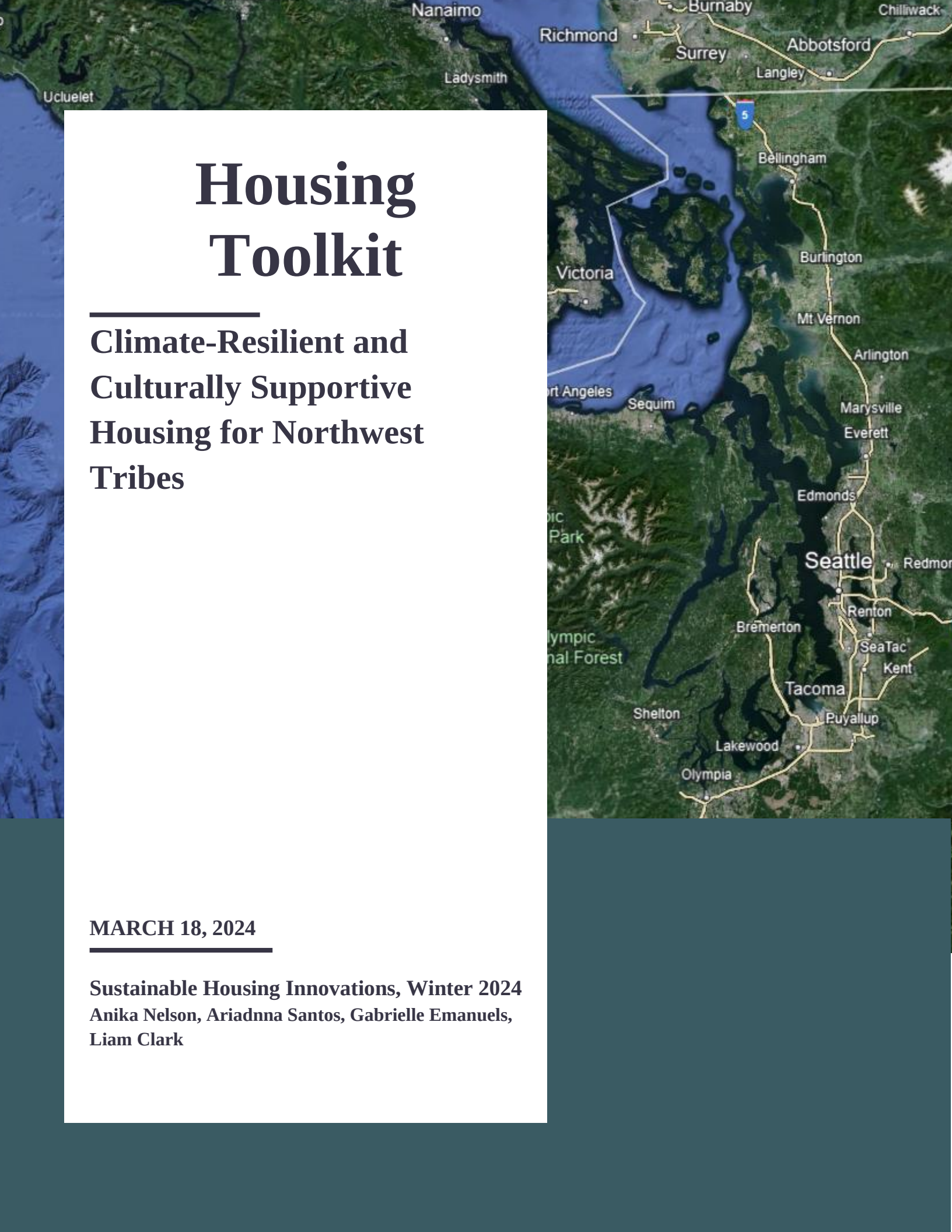




Housing Toolkit

Climate-Resilient and Culturally Supportive Housing for Northwest Tribes

MARCH 18, 2024

Sustainable Housing Innovations, Winter 2024
Anika Nelson, Ariadnna Santos, Gabrielle Emanuels,
Liam Clark

Executive Summary

The purpose of this toolkit is to provide a structure to developers designing and building housing for tribes in the Salish Sea region. The Salish Sea region is in a position to be significantly affected by climate change impacts, including increasing temperature extremes, flooding, coastal erosion, and forest fires. Since there is limited housing stock for tribes in the region and much of the current housing is not built to withstand the increasing severity of weather events, it is essential to increase the housing stock by building homes prepared for weather events to come. In addition to building climate-resilient homes, since these developments are focused on a specific cultural group, the homes should also support the tribes' customs and fit with their lifestyle. Therefore, one of the most critical components of the design process is hosting meetings with the community to ensure the design fits their needs. After a review of precedents and the community context, the proposed development is a cottage-style development of sustainable and climate-resilient homes oriented around a central green space that serves as the central gathering space for the neighborhood. The central climate-resilient features include storm-water preparation through rain gardens and limiting permeable surfaces, orienting the home to take advantage of sun exposure and breezes to regulate the home's temperature, and use of traditional western red cedar wood, which has stood up to the climate for centuries. Some key design features to increase the cultural supportiveness of the homes include the central community green space, the use of traditional materials such as cedar wood, and universal design to ensure that people of all ages and abilities can gather in the space. The proposed metrics for success are minimal needs for damage repair from weather and a reported increase in people's reported pride and comfort in their homes.

Purpose

This toolkit seeks to address the problem of the overall limited housing supply for tribal members in the Salish Sea region, where much of the existing housing was constructed without cultural specificity in mind, and homes are vulnerable to the impacts of climate change. In the coming years, the Salish Sea region is expected to experience increasing temperature extremes, flooding, soil erosion, and impacts from forest fires. We aim to address the effects of climate change through innovative housing design and traditional knowledge to prepare homeowners for the future effects of climate change. In addition, culturally supportive housing is vital for community development. The housing development process will hold tribal self-determination as a central value, and we aim to work with tribal communities to provide a housing solution that meets the needs of their community. Since housing is limited, it is crucial to start building more climate resilient housing since many homeowners are at risk of having to move out of their current home if there is a significant weather event and there currently isn't any housing for them to move to.

Literature Review

Designing climate-resilient and culturally supportive housing requires synthesizing knowledge from different sectors. Therefore, we reviewed the literature on design approaches for climate-resilient housing and elements of culturally supportive housing. According to our literature, the increasing severity of climate change is likely to disproportionately affect NW tribes located along the coast of the Salish Sea. We must address the effects of climate change in our housing design, including coastal erosion, flooding, temperature extremes, and forest fires. In a review of features of resilient housing, the authors, Ruiz and Mack-Vergara, concluded that the most resilient houses prioritized building on the safest land possible, using high-quality and local materials, and alternative sources for energy and water

(2023). In addition to being climate resilient, culturally supportive housing is critical to any tribal housing project. The article “Tribal Housing, Codesign, and Culturally Sovereignty” by Edmunds et al. focuses on the importance of putting tribal needs and wants at the forefront of design (2013). The article specifically focuses on the process of improving housing for the Pinoleville Pomo Nation (PPN) by putting cultural sovereignty at the forefront of the design process. The project included professors and students from UC Berkeley, as well as officials, employees, and citizens of PPN so that the people who would eventually be living in the housing had a strong influence on the design. The project also had a Native American planner as a senior advisor who could articulate the significance of specific design features to the other team members. The resulting developmental goals were to create housing that could accommodate tribal social life and cultural practices, be as off-grid as possible to reduce dependence on unreliable sources, be as personalized as possible, and be a source of pride for the

owners (Edmunds et al., 2013). Figure 1 shows that while designers on this project had to change and adapt certain parts of the housing design to appeal to local engineers and contractors, the fundamental design features could still be implemented thanks to the firm stance of cultural sovereignty first as seen. We intend to follow the lead of this project by honoring native voices and

traditional knowledge for the design of both culturally responsive and climate-resilient housing features. Another precedent is the Stillaguamish Village, built by 7 Directions Architects for the Stillaguamish Tribe in Arlington, Washington in figure 2. The project includes a wellness center, single-family homes, homes for elders, and a canoe shed. The designs incorporate many cultural elements that are shared among many tribes in the Salish Sea region, such as the use of cedar wood,

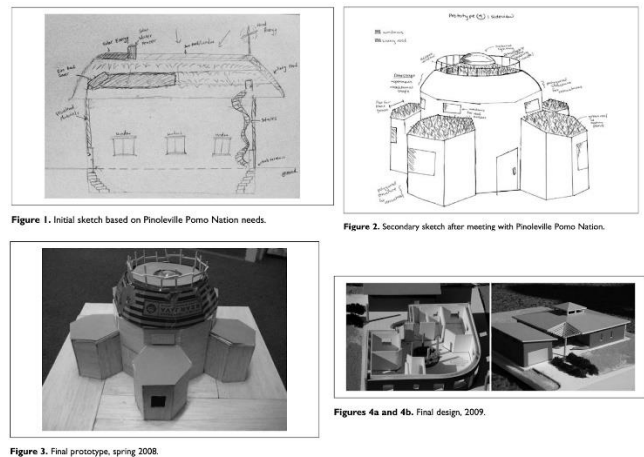


Figure 1: Evolution of design through community meetings (Edmunds et al., 2013)

buildings designed to resemble traditional plank homes, and space for canoe carving (7 Directions



Figure 2: Stillaguamish Village Masterplan by 7 Directions Architects

Architects, 2023). From the photos on the architect's website, it is also clear that they involved many tribal members in the design through multiple

community meetings where

people could work together and engage hands-on with the design (see Figures 3 and 4). Many climate-resilient and culturally supportive measures are intertwined for Northwest tribes since climate change will affect industries and elements essential to native identities, such as decreasing salmon populations (Dalton, M.M., P.W. Mote, and A.K. Snover, 2013).

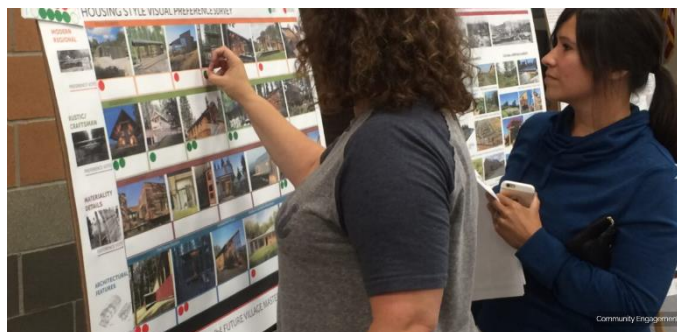


Figure 3: Community Meeting on the Designs of Stillaguamish Village (7 Directions Architects, 2023)



Figure 4: Community members hands-on with the proposed design (7 Directions Architects, 2023)

From our literature review, we concluded that our process for designing climate-resilient and culturally supportive housing must place native voices at the forefront and hold many community meetings for input before establishing a final design. In addition, we will look to precedents of homes built by other tribes in the Salish Sea region for examples of climate-resilient features, such as the designs from 7 Directions Architects. Based on the review, features that we would likely include in our design proposal include flood water mitigation, self-sufficient energy systems, and locally sourced

materials. There is still room for more research on elements of Northwest homes since one gap in the research is that many of the articles focused on tribes in other regions, so not all of the findings apply to the Salish Sea region. Additionally, most articles addressed either climate-resilient or culturally supportive features rather than the importance of their intersection when designing homes for tribal communities.

Community Context

To create a framework for our process, we chose an example site to illustrate the steps to developing our model for climate-resilient and culturally supportive housing. Our proposed site is on Gooseberry Point on the Lummi Nation reservation (see Figure 5) and would, therefore, serve Lummi Nation tribal members. The first step after choosing a site was to complete a community assessment to have context to inform our design suggestions for the site. This was a rapid assessment of the context and, in application, should also include speaking to community members directly to gather a better understanding than the statistics alone provide. The Lummi enrollment office reports that there are currently 5,258 enrolled members, with 2,920 members living within the reservation boundaries, 959 members living off the reservation but still in Whatcom County, and 1,379 members living outside

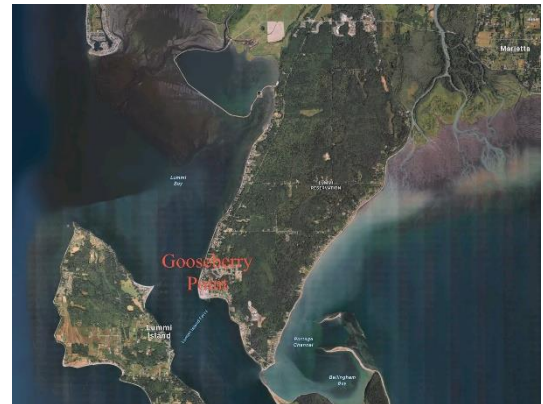


Figure 5: Satellite view of Lummi Peninsula and Gooseberry Point. Photo from Apple Maps

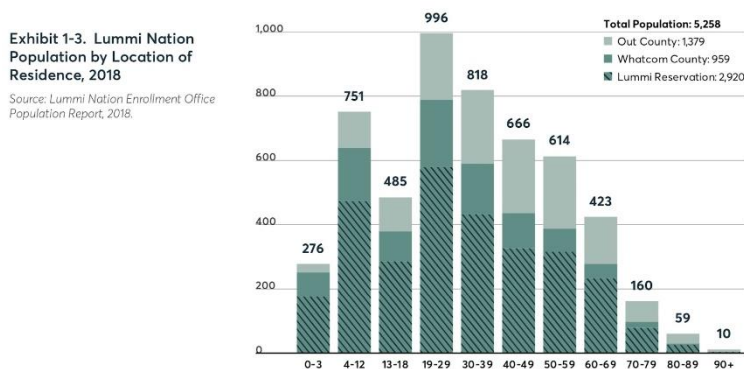


Figure 6: Age distribution of Lummi Nation population (LIBC Planning Department, 2022)

Whatcom County (LIBC Planning Department, 2022). The census for Lummi reservation lands reports that the total population is 5,211, including tribal and non-tribal members, the median age is 40.2, and there are just shy of 1000 people 65+ living there (CNMP, US

Census Bureau, n.d.). See Figure 6 for a more detailed age breakdown of the population. The largest employment sectors are education, healthcare, and social services (CNMP, US Census Bureau, n.d.), and Lummi Nation is the second largest employer in Whatcom County (US Census Bureau, n.d.).

However, while Lummi is a major employer, the median annual household income on Lummi Reservation is about \$67,414 (Census Reporter, n.d.), which is about \$12,000 lower than the median annual income in Whatcom County of \$79,200 (US Census Bureau, n.d.). 13.2% of people on Lummi Reservation are under the poverty line, with 11.8% of people 65+, 12.6% of people under 18, and 6.9% of families under the national poverty line (CNMP, US Census Bureau, n.d.).

In terms of housing, Lummi has 2,270 housing units, with 1,921 occupied and 349 vacant. Of the occupied units, 1,414 are owner-occupied, and 507 are renter-occupied. Lummi has a homeowners

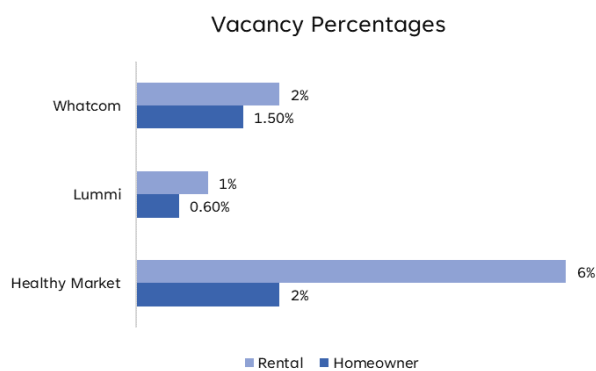


Figure 7: Vacancy rates compared between Whatcom County, Lummi Nation, and a healthy market for homeowners and renters. Graph by Anika Nelson with data from US Census Bureau

vacancy rate of 0.6% and a rental vacancy rate of 1.0% (CNMP, US Census Bureau, n.d.), which falls well below the healthy vacancy rate of around 2% for homeowners and 5-7% for rental units (see Figure 7) (City of Bellingham, 2022).

The average household size for Lummi is 2.59 for

owner-occupied units and 2.79 for rental units (CNMP, US Census Bureau, n.d.). In comparison, the average household size in Whatcom County is 2.43 (US Census Bureau, n.d.), which may be an indication of cultural differences where, for Lummi, extended family members are more likely to live together or that sharing is economically necessary since the annual incomes for Lummi are significantly lower than in Whatcom County and there is more limited housing stock. The median home price in Lummi is \$358,100, with 585 owners having a mortgage with an average monthly cost of \$1,984 and 829 without a mortgage and monthly expenses of around \$562. The median rental price is \$742 per month. 4,668 people of the 5,211 population live in the same house they did a year ago. Still, about 500 people moved to the Lummi reservation from

within Whatcom County or the United States, which signifies a need for more housing (CNMP, US Census Bureau, n.d.). The CLUP reports that the typical

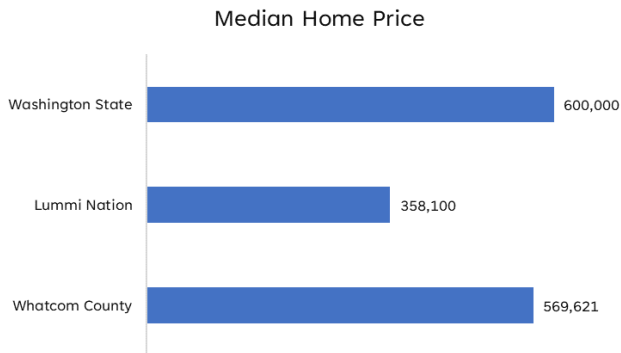


Figure 8: Median home price compared between Washington State, Lummi Nation, and Whatcom County. Chart by Anika Nelson with data from US Census Bureau

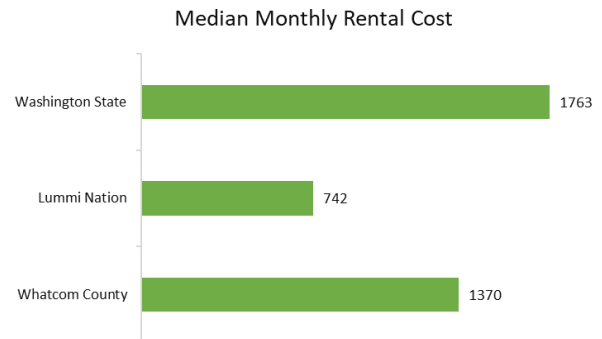


Figure 9: Median monthly rental cost compared between Washington State, Lummi Nation, and Whatcom County. Chart by Anika Nelson with data from US Census Bureau

housing occupied by Lummi tribal members is a new or used manufactured mobile home or 30-50-year-old government housing built below standard and has not been adequately maintained or inspected, leading many of the properties to fail to provide proper shelter (LIBC Planning Department, 2022). Takeaways from Lummi community meetings in the Lummi Community Planning Document show that Lummi members strongly felt the need for diverse housing types at affordable prices and close to transportation, housing for elders to age in place, and housing that preserves and strengthens their culture. They also expressed concerns over the quality of current housing units (LIBC Planning Department, 2022). Lummi also created the Lummi Nation Housing Authority (LNHA) in 2006 to remedy unsafe housing conditions and provide employment through improving such homes and constructing new homes. LNHA manages over 350 housing units, including duplexes, triplexes, and small single-family homes. The Lummi community is open to multi-family housing as the current home prices on the Lummi Peninsula are between \$265,000 and \$500,000, which is not affordable for most tribal members (LIBC Planning Department, 2022). The average single-family home is around 1000 square feet on around ¼ acre, built between 1965 and 1985.

Within the Gooseberry Point area, there appear to be primarily single-family homes, but many have a separate garage or outbuilding that could be functioning as an extra unit. Gooseberry Point has little to no mixed-use development since it is mainly residential and has few storefronts besides local fishermen, including lacking access to a grocery store within the reservation boundaries.

NW tribes are situated in the Puget Lowlands ecoregion, where its proximity to the Pacific Ocean gives this area a mild maritime climate. For Lummi, 43% of their reservation are wetlands entirely (see Figure 10), while 50% are covered in forests (see Figure 11)

(LIBC Planning Department, 2022). As the density of neighborhoods grows, units tend to be placed near the coastlines; notable residential neighborhoods include Gooseberry Point and Neptune Beach. However, as sea levels rise and climates change, these properties will be threatened with erosion and flooding. By 2080s, sea-level rise will accelerate from 0.5 meters to 1 meter above present-day levels, while stream runoffs will increase from 20% to 30% by 2040s (Grossman et al., 2023). The Nooksack River has a long history of extreme flooding events, such as the February 2020 Super Bowl Flood, which totaled \$2.5 million in damages to public infrastructure and \$1.6 million to businesses and homes (Mittendorf, 2020). The Lummi Indian Business Council also notes that the peninsula south of the Nooksack and Red River is isolated yearly from emergency services such as fire

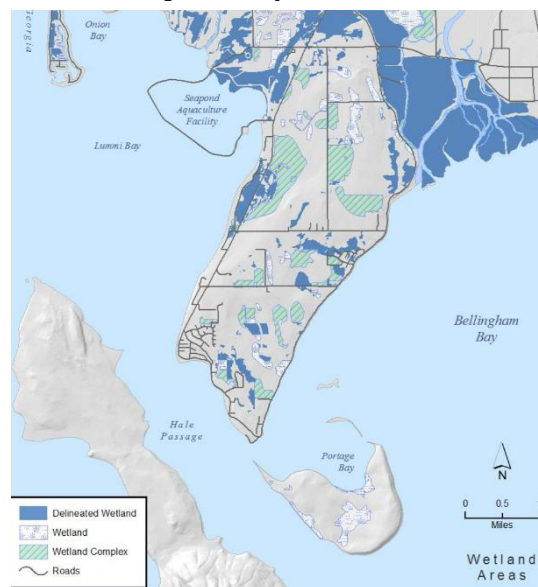
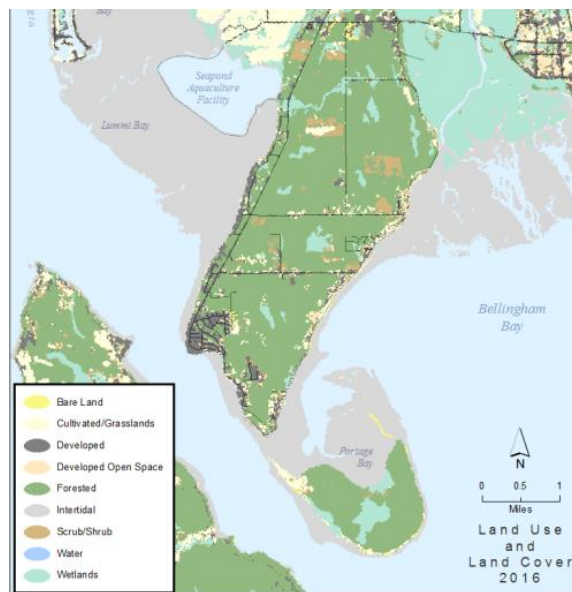


Figure 10 (above): Map of wetland areas on Lummi Peninsula (Lummi Indian Business Council, 2020)

Figure 11(below): Map of Land Use and land cover on Lummi Peninsula (Lummi Indian Business Council, 2020)



departments and ambulances due to flooding. Lummi Nation plans to combat this by using vegetation and green engineering on riverbanks and concentrating development in already-developed areas to encourage a mix of building types and diverse housing. While the ecology differs throughout the Salish Sea region, coastal erosion and flooding are common concerns to most land along the coast.

There is a high need for affordable housing, upgrading and modernizing houses, and creating more diverse housing units. Unfortunately, due to historically unjust government policies, parcels are owned by multiple people. At least 51% of parcel owners must consent to any development on said land - however, many owners cannot be located, which makes the land unusable. This does not bode well for Lummi's need to create more units - 1,800 new units and 200 replacements are needed in the upcoming 20 years (Lummi Indian Business Council et al., 2023). Indigenous tribes are also disproportionately affected by substandard living. Lummi Nation reports that 40% of on-reservation housing is considered sub-standard, with $\frac{1}{3}$ of homes being overcrowded, less than $\frac{1}{2}$ of homes being connected to the public sewer system, and 16% lacking indoor plumbing. While the median home value for a home in the Lummi Reservation is about 58.7% lower than the median home value for the State of Washington, most of these houses are not affordable to most tribal members. Average household size in NW tribes also tends to be higher than in Washington State due to the tendency of tribal households to be multigenerational, which calls for more multi-family homes to be generated in the near future.

Stakeholders in Lummi Nation advocate for housing availability, preserving culture, and combatting homelessness. These stakeholders are individuals from multiple groups that support the Lummi tribe, such as the Lummi Nation Housing Authority, the Lummi Community Development Financial Institution, and the Tribal Employment Rights Office (*LIBC* Planning Department, 2022). Housing projects occurring on native land, such as the Sche'lang'en Village, can also be funded with loans from HUD, though it must be noted that projects such as these must meet specific standards to receive government funding.

The Lummi have put resources into tribal housing through developments by the Lummi Housing Authority and have also been updating the infrastructure on the reservation. They recently built a new clinic (see Figure 12) that consists of a medical and dental clinic, pharmacy, imaging facilities, a childcare center, a behavioral health clinic, and a garden with medicinal plants. It also serves as a model for sustainable building practices since it uses solar power and rainwater irrigation for the garden, which are features we aim to include in our housing development. In addition, the design



Figure 12: Model of the newly built Lummi Tribal Health Center located on the northern part of the reservation. Rendering by kmd Architects

incorporated traditional building practices and was constructed to mimic a traditional Lummi longhouse. Being a multi-functional community center, the clinic is a functional piece of infrastructure and a physical amenity for the community. (Lummi Nation Wellness

– kmd

architects, n.d).

Our proposed

site is also down the road from a pre-k childcare facility and a 5-minute drive or 20-minute walk to the Lummi Tribal School. Our site is also within a five-minute drive from two parks, Hermosa Beach and West Beach, and is adjacent to protected wetlands. The Lummi Nation Stommish grounds and the Lummi ROC community center are both part of the Gooseberry Point neighborhood and a crucial part of the neighborhood fabric (see map in Figure 13). While the neighborhood provides cultural support systems, the Lummi reservation is classified by the

USDA as a “food desert” where there is limited access to fresh produce and other healthy foods,

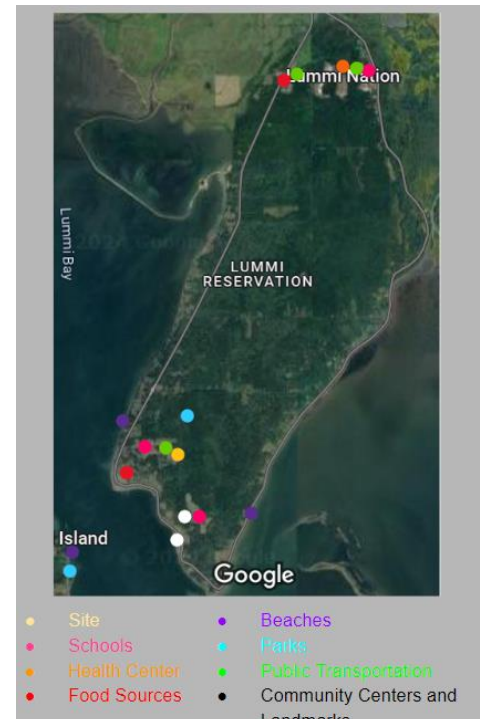


Figure 13: Map of services and their proximity to the proposed site by Liam Clark with location data from Google Maps

leading to higher levels of health problems in the community (*LIBC Planning Department, 2022*). The nearest grocery store is a 20-minute drive away and outside of walking distance, and public transportation would take roughly an hour and 20 minutes round trip, which isn't feasible for most people. Another concern is access to emergency services since, as noted earlier, flooding can isolate homes from emergency services. Since our homes are intended to serve a broad range of ages, including the elderly, who are most susceptible to injury and illness, timely access to medical care is essential. The Lummi Tribal Health Center is about a 9-minute drive, but anything more serious would require a trip to St. Joseph's Hospital in Bellingham, which is about 25 minutes away.

The U.S. Department of Housing and Urban Development, commonly known as HUD, sets the majority of policies for development within the states and works closely with tribal reservations to set policies and regulations for land development and building codes. HUD's main programs involve public and subsidized housing, mortgage and loan insurance, managing development grants, and tribal consultations (*Q and A about Hud. HUD, 2021*). HUD Tribal Consultation works with tribes when developing policies and regulations that affect federally recognized tribes (*Hud Tribal Consultation, n.d.*). The Native American Housing Assistance and Self Determination Act of 1996, NAHASDA, manages housing assistance for native tribes and controls the block grant program (NAHASDA, *n.d.*). As we look for funding options, the most notable are the Indian Housing Block Grant for affordable housing developments and the Indian Community Development Block Grant, which helps finance community facilities, land acquisition, and infrastructure facilities. The Section 184 Indian Home Loan Guarantee Program is notable as it aims to increase native home ownership by making home loans more accessible (*Resources for tribal housing & TDHES, n.d.*). Federally recognized tribes can also buy and sell land through trust or fee lands. Trust lands are land sold to the government by a tribe for the government to hold in trust so that it may be developed for the economic benefit of the tribe (*Office of Trust Services, n.d.*). *Fee lands* are no longer in trust and can be sold to anyone. Often, the land is

held by non-tribal members. The General Allotment Act of 1887, or the Dawes Act, gave parcels of land to individual tribe members, but after 25 years, the lands were taxed under state law, and many native landowners could no longer afford them. This reduced Native land ownership and still affects tribes today (*Native American lands: Ownership and governance*, n.d.). Therefore, selecting a parcel of land must begin with determining who owns the land and if it can be purchased by either a developer or the tribe. In the case of our proposed site, the Lummi Nation Housing Authority is already building adjacent and owns the land, so we can proceed to the next step of establishing funding for the project. There are many different government-supported grants and funds available, but sometimes, the regulations on how the funds are used or what they can be used for are a hindrance, so the next step is to identify which funding options are applicable to our proposed development.

In addition to the federal government, there are Lummi Tribal codes based on the Lummi Tribal Constitution. The 2006 Lummi Tribal Constitution states in Article VI that the Lummi business council may purchase and lease land for tribal use (*Constitution and Bylaws of the Lummi Tribe of the Lummi Reservation, Washington as Amended*, 2016). Additionally, within the Housing Authority Code Title 37, chapter 37.07 states that exceptions to the established building and housing codes can be made for projects that are “reasonable and necessary to promote economy and efficiency in the development and operation of any project, and at the same time safeguard health and safety” (Lummi Nation Housing Authority, 2006). This also includes making changes to the zoning of the site and surrounding territory, if necessary, which applies to our proposed development since it is currently zoned as “forested”, but as the LNHA is already building in the area, changing the zoning of the site would likely be approved. The Lummi Nation Housing Authority can also limit admission to low-income residents for particular residences (Lummi Nation Housing Authority, 2006).

Problem Statement

There is an overall limited housing supply for tribal members in the Salish Sea region, and much of the existing housing is not constructed with cultural specificity in mind. Historically, tribes have not had the resources to develop culturally supportive and climate-resilient housing, and many tribes are in regions vulnerable to the impacts of climate change.

Principles and Patterns

We developed our guiding principles through research of tribal values with a heavy emphasis on the Lummi Nation since our example case is on Lummi Nation land for tribal members. Through our review of tribal community plans within the Salish Sea region, we settled on three guiding principles. The first principle is “sovereignty” since control over one’s choices and the ability to self-govern is a right for tribes that has historically been stripped by the colonizing government. Having control over something as fundamental as housing is essential to protecting the Indigenous culture. The second principle is “community” because the Lummi, along with many other northwest tribes, place great importance on their community and look to the wisdom of elders and the energy of youths to strengthen their community. The final principle is “Conservation” since native tribes have always lived with and been caretakers of the land. Hence, the conservation of the land they live on is not only vital to sustainability but also to protecting their cultural connection to their traditional lands.

Figure 14 demonstrates how the principles fit together to create a cohesive design strategy. The principles of “Sovereignty,” “Community,” and “Conservation” led to the development of our seven central patterns, which are prioritizing native and established vegetation, centering community spaces,

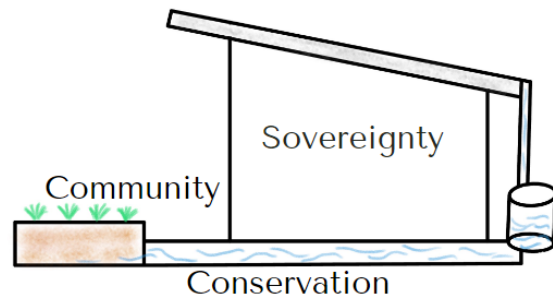


Figure 14: Model of how the principles will inform the patterns of design. Model by Anika Nelson

locally sourced materials, responsible water usage, climate change preparation, natural sunlight and ventilation, and universal design. The pattern of “prioritizing native and established vegetation” is drawn from the principle of conservation since using native plants is best for the soil and overall ecology of the area, as well as creating a sense of place for the residents since the native plants may also include traditional medicinal plants. “Centering community spaces” is clearly drawn from the principle of community and having a central gathering area is a way to facilitate engagement between neighbors. The pattern of “locally sourced materials” is drawn from both the principles of sovereignty and conservation. We plan to prioritize giving back to the community by supporting local businesses owned by tribal members, such as lumber from responsibly managed forests operated by tribal



Figure 17: Locally Sourced Materials. Drawing by Gabrielle Emanuels

members. Using local materials also reduces the carbon footprint of developing the homes since materials don’t have to be transported as far. “Responsible water usage,” “climate change preparation,” and “natural sunlight and ventilation” are all patterns derived from the principle of conservation since one of the main goals of the project is to increase the climate resilience of the homes and therefore also increase the sustainability of homes. The specific features of these



Figures 15 and 16: Prioritizing Native and Established Vegetation (above) and Centering Community Spaces (below). Drawings by Gabrielle Emanuels

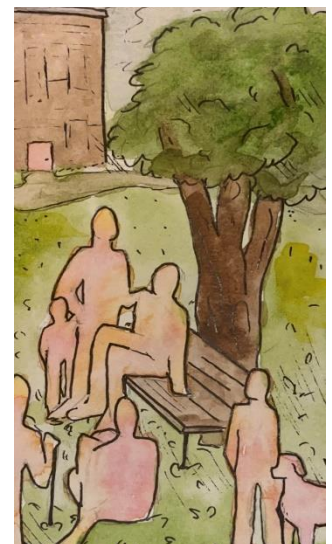




Figure 18: Responsible Water Usage. Drawing by Gabrielle Emanuels

patterns we intend to include in our housing model will be discussed later when we introduce possible housing configurations and designs. Finally, “universal design” is a pattern that supports the principle of community since, with accessible designed spaces, all people, such as kids, elders, and people with disabilities, can be included in the physical space. Figures 15 through 21 demonstrate the patterns visually.



Figures 19-21: Climate Change Preparation (left), Natural Sunlight and Ventilation (center), and Universal Design (right). Drawings by Gabrielle Emanuels

Approach to the Housing Model

Our approach to the housing model is to provide a framework for housing developers, either private or tribal, to follow when designing homes for tribal members in the Salish Sea region. The model aims to provide developers with design ideas for climate-resilient and culturally supportive housing through an example development. As tribal needs vary throughout the region and there is no one-size-fits-all approach, our housing model is intended to provide ideas to developers, who may choose which features are the best fit for their site. In creating this framework, we struggled to find many precedents for climate-resilient tribal housing in the Northwest, which limited our ability to compare different design styles and approaches. There were many good precedents in other regions of

the United States and abroad, but due to the cultural specificity of tribal housing and differing climates, the precedents were not as applicable to our design suggestions. The design suggestions are also subject to change since one of the most critical parts of the development process that we want to encourage is to put tribal voices at the forefront of the design process. Another difficulty was finding solid information on the building codes and policies for the tribes. However, this is mostly positive since the tribal authorities have more leeway to change the zoning and other codes than working with a city government.

Developers using this toolkit should survey homeowners on their feelings of cultural supportiveness provided by the home and the home's tolerance to weather events to ensure they have succeeded in providing homes that are both climate resilient and culturally supportive to tribal communities in the Northwest. While reported feelings of cultural supportiveness are more subjective, measuring a home's ability to withstand weather events can be done through multiple means, such as tracking repairs over a period of time.

Configurations and Design

The main foundations of our design configurations come from our patterns and from reading the most recent Lummi Nation Community Planning document from 2022. As a caveat, if the homes are not intended for Lummi, then the community plan should be substituted for the tribe in question. However, in our research, some of the tribes had far more robust community plans than others, so it may still be beneficial to look at the community plans of other tribes nearby, especially for non-tribal-specific information such as ecology information. We first decided that our proposed home would be single-family based on the

Exhibit 1-7. What is the right type of future growth for the following areas?

Source: Lummi Community Survey, 2019.



Figure 22: Table of responses from community meeting on what types of development people want to see in Gooseberry Point. (LIBC Planning Department, 2022)

responses in Figure 22, which show that most respondents preferred Gooseberry Point development to focus on single-family homes. The following steps were to implement the patterns into our design, which is how we arrived at the typology of cottage-style homes around a central community



Figure 23: Example of a cottage-style neighborhood called 'Pocket Neighborhood' by Ross Chapin Architects <https://pocket-neighborhoods.net/examples/single1.html>

affordable to more people. In deciding on a layout for cottage homes, we looked to precedents from Ross Chapin Architects, who have created many cottage-style neighborhoods throughout the Pacific Northwest, as shown in Figures 23 and 24 (*Pocket Neighborhoods • Creating Small Scale Community in a Large Scale World*, n.d.). We also looked at the designs of the

greenspace. Cottage homes provide the best balance between shared green space and personal private space, and since they are smaller, they are

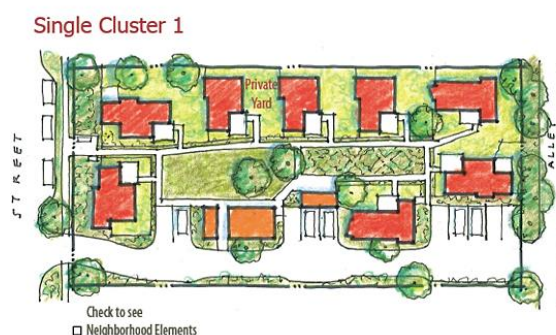


Figure 24: Example layout for a cottage-development by Ross Chapin Architects <https://pocket-neighborhoods.net/examples/single1.html>

included the single-family development where the homes are positioned around a central park and playground in Figure 25 (7 Directions Architects, 2023). The garden and open space in the central ring



Figure 25: Homes oriented around a green space and playground in the Stillaguamish single-family development (7 Directions Architects, 2023)

between the cottage homes are as essential to our designs as the physical homes. The open space will include open space for community gatherings, a garden for growing food and traditional medicinal plants, and a space to display art from native artists. The goal of the

park area being the center of the community is to create an inviting and accessible space that

encourages neighbors to interact and form connections. The garden also integrates with the water recycling initiatives since the garden will be irrigated by the rainwater collected from the roofs of the



Figure 26: Diagram of a rain garden (Not Just Another Pretty Flower: The Basics of Rain Gardens, n.d.)

family homes. Separate rainwater gardens can also be implemented around the property, such as on the outer ring, since they provide a place for stormwater drainage.

As shown in Figure 26, a rain garden is a depressed area that collects water runoff from roofs, roads, and other impervious surfaces and allows rainwater to soak into the

ground and irrigate plants in the depressed area (*Soak up the Rain: Rain Gardens* | US EPA, 2023).

Stormwater drainage is essential for preparing for flooding. If rainwater is directed into the rain gardens through gravity, it will be less likely to soak in around the foundations of the homes and cause home damage. In addition to rainwater collection and rain gardens, limiting the number of impermeable surfaces allows water to naturally soak into the ground and prevent further problems in the watershed (Frazer, 2005). One alternative to traditional

paving is permeable pavements that allow water to run through them rather than diverting the water that falls on them (Frazer, 2005). Using permeable pavement like in

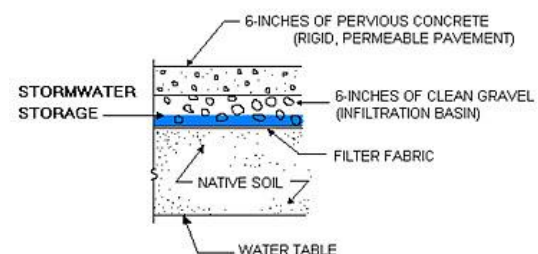


Figure 27: Diagram of the structure of permeable pavement. (California n.d.)

Figure 27 to pave the walking paths would allow for proper water drainage while providing a flat and stable walking surface for people with mobility issues (California, n.d.).

As climate change continues to push temperature extremes, we need to consider features to not only warm the home in the winter but also keep it cool in the summer. One way is to use higher-grade insulation in the home's walls and roof, double pane windows, and window coverings to reflect sunlight during the summer. However, these coverings can be opened to allow natural light during the

dark winter months. Another building system that helps maximize energy efficiency is orienting the home on the site to take advantage of natural breezes and sunlight. The home design in Figure 28

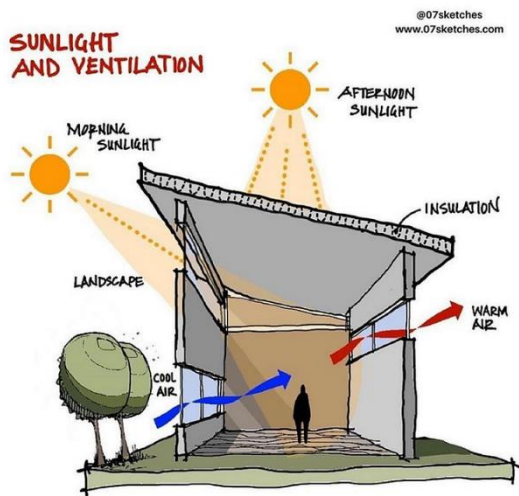


Figure 28: Diagram of a system of passive cooling by 07Sketches *Ventilation systems in Architectural design* | by HighArchi | Medium

shows how the home shape can allow natural ventilation through strategically placed windows and orienting the home for the best sun exposure. Many homes in the Salish Sea region receive steady breezes off the water, so positioning the windows to receive the best breezes will help cool the homes as the summer temperatures get more extreme. Sun can also be used to the homeowners' advantage through the installation of solar panels.

Although western Washington is often rainy and cloudy, solar panels can still generate plenty of energy and be a viable energy source (Hewitt, 2023). Around 90% of Washington's energy production is renewable, so in terms of sustainability, adding solar to housing is unnecessary for conservation (*U.S. Energy Information Administration - EIA - Independent Statistics And Analysis*, n.d.). However, solar energy would increase tribal energy sovereignty, making tribes less reliant on services, which can be patchy and expensive on many reservations.

Using local materials is essential for both sustainability and cultural supportiveness. For the Lummi Nation and many other tribes in the Salish Sea regions, the use of cedar wood dates back to early plank homes such as the one depicted in Figure 29 (Wallace, 2017). Cedar is still an excellent building material since it is durable and holds up well in damp climates. Purchasing cedar wood locally will support local businesses and decrease the energy needed to transport the lumber to the building site. Another



Figure 29: An artist sketch of a shed roof plank house, circa 1790 (Wallace, 2017)

consideration would be using prefabricated homes such as those designed and built by Lindal Cedar Homes in figures (!&!) that use western red cedar from responsibly managed forests (Lindal Cedar Homes, 2024). Prefabricated homes can help speed up the permitting process and fit with many of our



Figure 30: "Cedar" - 3 bedrooms (Lindal Cedar Homes, 2024)



Figure 31: "Olive" - 2 bedrooms (Lindal Cedar Homes, 2024)

design patterns. In addition, Lindal allows customization of their homes so the home can be tailored to best fit the location. Another opportunity would be to save space on the home plots so homeowners can add their own ADU in the future. The ADU could be another living unit or could provide another function, such as boat storage since fishing is a large industry and part of the culture of most Salish Sea tribes.

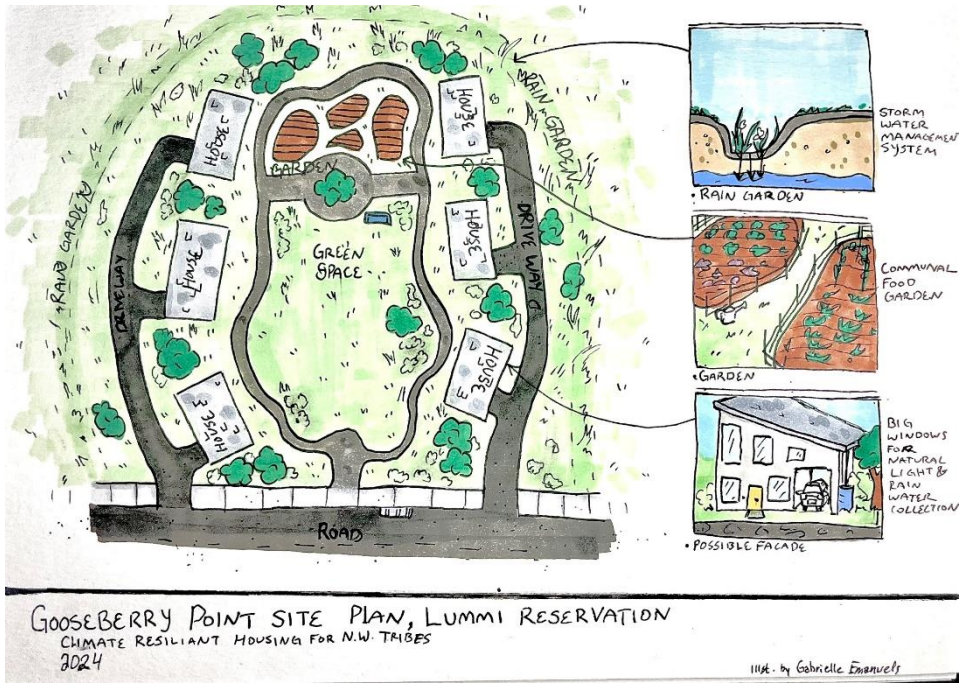


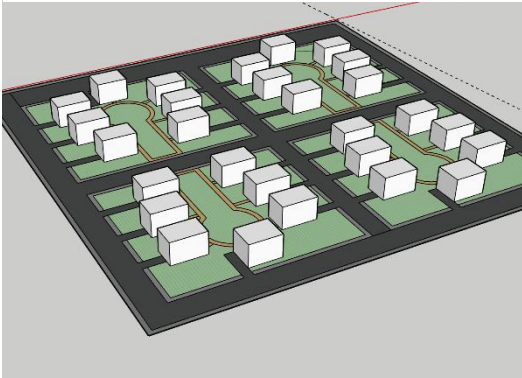
Figure 32: Drawing of potential site design by Gabrielle Emanuels

After gathering research on the many features available to make the homes as climate resilient as possible while also creating a culturally supportive



Figure 33 (above): Overlay of design on proposed site in Gooseberry Point. Rendering by Ariadnna Santos

Figure 34 (below): Example of how development can be scaled up. Model by Ariadnna Santos



environment, we combined the ideas into the drawing in Figure 32 and then a SketchUp model of the basic design and how it fits onto the proposed development site (Figure 33). Figure 34 shows how the design could be scaled up to create a larger development composed of small communities. As development scales, one way to build community is to prioritize keeping extended families within the same area to promote traditional tribal kinship.

The next step would be to host another community input meeting with tribal community members and key stakeholders to revise the design and ensure that it meets the required building guidelines.

Policy Considerations

The Lummi Tribal codes are based on the Lummi Tribal Constitution and enforced by the Lummi Housing Authority. The 2006 Lummi Tribal Constitution states in Article VI that the Lummi business council may purchase and lease land for tribal use (*Constitution and Bylaws of the Lummi Tribe of the Lummi Reservation, Washington as Amended*, 2016). Additionally, within the Housing Authority Code Title 37, chapter 37.07 states that exceptions to the established building and housing codes can be made for projects that are "reasonable and necessary to promote economy and efficiency in the development and operation of any project, and at the same time safeguard health and safety" (Lummi Nation Housing Authority, 2006). This also includes making changes to the zoning of the site

and surrounding territory if necessary. The Lummi Nation Housing Authority can also limit housing project admission to low-income residents (Lummi Nation Housing Authority, 2006). We would also have to consider Lummi's zoning codes, especially ones that deal with flooding regulations like Title 15A from the Lummi Nation Code of Laws. According to the title, the design must be centered around minimizing flood damage. This means additions such as new constructions needing to be anchored to the ground and utilities such as electricity and heating needed to be elevated to prevent water damage, which can be easily added to our project.

In addition to tribal building codes, policies from HUD and the International Rescue Committee are restraints that must be considered. Most of these codes are written in HUD's chapters on general acceptability criteria, building design, and site design. 302-2 Noise Control, 402-2 Handrails and Railings, 204-1 Streets, and 303-2 Building Parking Setback Distance are all key codes that inform our design and enhance the safety and well-being of residents (1994). IRC codes are implemented to protect residents and make access to emergency services quicker in case of emergencies. Examples are R304: Floor/Room Area, R802.11.1 Uplift Resistance, R317.1.4: Columns and Posts, and R317.1: Wood in Contact with Concrete or Masonry (International Code Council, 2021).

While these codes may be extra considerations for our current project, they guide our designs to help produce housing that is specific to our current site proposition and accessible for all. Compliance with HUD, the International Rescue Committee, and Lummi's Code of Laws informs us about our design's safety needs. Certain articles and laws, such as Title 37 from Lummi Nation's Housing Authority Code, can help us bypass certain zoning and code laws and focus on low-income families who need housing the most.

Metrics

We propose to increase housing stock for tribal members by constructing new homes that are climate-resilient and culturally supportive. The suspected outcome is that the new construction of homes will increase the housing vacancy rate as more homes are available, and the homes will be able to stand up to weather events so that housing stock is less affected by climate change. The first metric we will use to measure success is to track the housing vacancy rates. A healthy vacancy rate is about 2% for homeowners, so if the housing vacancy rate increases after the construction of new homes, the housing stock can better support the community. Secondly, we will measure the success of climate resiliency by whether or not homes sustain significant damage that requires homeowners to move out for repairs in the next 30 years. If the home only experiences minor repair needs from climate events over the next 30 years, the home has succeeded. Cultural supportiveness is the most complicated factor to measure. Still, one idea is to measure the satisfaction and pride of tribal members in their homes before and after moving into the home.

Bibliography

California, S. O. (n.d.). *Erosion Control toolbar: Permeable Paving* | Caltrans.

<https://dot.ca.gov/programs/design/lap-erosion-control-design/tool-1-lap-erosion-control-toolbox/tool-1nn-40-permeable-paving>

Center for New Media & Promotion (CNMP), US Census Bureau. (n.d.). *My tribal area*.

<https://www.census.gov/tribal/?st=53&aianihh=2070>

Census Reporter. (n.d.). *Census profile: Lummi Reservation CCD, Whatcom County, WA*.

<https://censusreporter.org/profiles/06000US5307391760-lummi-reservation-ccd-whatcom-county-wa/>

Chapter 2 General Acceptability Criteria . U.S. Department of Housing and Urban Development (HUD).

(1994a). <https://www.hud.gov/sites/documents/49303GC2GUID.PDF>

Chapter 3 Site Design . U.S. Department of Housing and Urban Development (HUD). (1994b).

<https://www.hud.gov/>

Chapter 4 Building Design. U.S. Department of Housing and Urban Development (HUD). (1994c).

<https://www.hud.gov/sites/documents/49101C4HSGH.PDF>

City of Bellingham. (2022, November 1). *Housing FAQs - City of Bellingham*.

<https://cob.org/services/planning/key-housing-questions>

Constitution and Bylaws of the Lummi Tribe of the Lummi Reservation, Washington as Amended.

Lummi Nation. (2016, February 1). <https://www.lummi-nsn.gov/>

Dalton, M.M., P.W. Mote, and A.K. Snover. (2013). *Climate Change in the Northwest: Implications for Our Landscapes, Waters, and Communities*. Washington, DC: Island Press.

[climatechangeinthenorthwest.pdf \(pnwcirc.org\)](https://pnwcirc.org/climatechangeinthenorthwest.pdf)

Edmunds, D. S., Shelby, R., James, A., Steele, L., Baker, M., Perez, Y. V., & TallBear, K. (2013).

Tribal Housing, codesign, and cultural sovereignty. *Science, Technology, & Human Values*, 38(6), 801–828. <https://doi.org/10.1177/0162243913490812>

Frazer L. (2005). Paving paradise: the peril of impervious surfaces. *Environmental health perspectives*, 113(7), A456–A462. <https://doi.org/10.1289/ehp.113-a456>

Grossman, E. E., vanArendonk, N. R., & Nederhoff, K. (2013). Compound flood model for the Lower Nooksack River and Delta, Western Washington—Assessment of vulnerability and nature-based adaptation opportunities to mitigate higher sea level and stream flooding. *Scientific Investigations Report*. <https://doi.org/10.3133/sir20235047>

Hewitt, S. (2023, August 22). Going solar can work, even in the cloudy Pacific Northwest — but do the math, one homeowner warns. *The Columbian*.

<https://www.columbian.com/news/2023/aug/19/going-solar-can-work-even-in-the-cloudy-pacific-northwest-but-do-the-math-one-homeowner-warns/>

Hud Tribal Consultation. HUD.gov / U.S. Department of Housing and Urban Development (HUD). (n.d.-a). <https://www.hud.gov/TribalConsultation>

International Code Council. (2021a). Chapter 3 Building Planning. 2021 WASHINGTON STATE RESIDENTIAL CODE | ICC DIGITAL CODES. <https://codes.iccsafe.org/content/WARC2021P1/chapter-3-building-planning>

International Code Council. (2021b). Chapter 8 Roof Ceiling Construction. 2021 WASHINGTON STATE RESIDENTIAL CODE | ICC DIGITAL CODES. <https://codes.iccsafe.org/content/WARC2021P1/chapter-8-roof-ceiling-construction>

Lindal Cedar Homes. (2024, March 13). *Lindal Cedar Homes | Build your beautiful custom home with us*. <https://lindal.com/>

Lummi Indian Business Council. (2020). *Lummi Nation Atlas: An overview of the history, natural and economic resources, and government of Lummi Nation*. Retrieved February 20, 2024, from https://www.lummi-nsn.gov/userfiles/592_2020LummiAtlas.pdf

Lummi Indian Business Council & Lummi Commercial Company. (2023). Comprehensive Economic Development Strategy. *Lummi Nation Comprehensive Economic Development Strategy*.

Lummi Indian Business Council Planning Department. (2022). Lummi Nation Community Plan. https://www.lummi-nsn.gov/userfiles/1034_LummiCommunityPlanFinalDraft20221127-compressed.pdf

Lummi Nation Housing Authority. (2006). *TITLE 37 Lummi Nation Code of Laws Housing Authority Code*. https://www.lummi-nsn.gov/userfiles/393_Title37HousingAuthorityCode.pdf

Lummi Nation Wellness — *kmd architects*. (n.d.). Kmd Architects.

<https://www.kmdarchitects.com/lummi-nation-wellness>

Mittendorf, R. (2020, February 21). *Whatcom Super Bowl weekend flood damage in millions, federal aid sought*. Bellingham Herald. <https://www.bellinghamherald.com/news/weather-news/article240514381.html>

NAHASDA. (n.d.). HUD.gov / U.S. Department of Housing And Urban Development (HUD). https://www.hud.gov/program_offices/public_indian_housing/ih/codetalk/nahasda

Native American lands: Ownership and governance. Native American Lands | Ownership and Governance | Natural Resources Revenue Data. (n.d.). <https://revenuedata.doi.gov/how-revenue-works/native-american-ownership-governance/#>

Not just another pretty flower: the basics of rain gardens. (n.d.). The National Environmental Education Foundation (NEEF). <https://www.neefusa.org/story/environmental-education/not-just-another-pretty-flower-basics-rain-gardens>

Office of Trust Services. Office of Trust Services | Indian Affairs. (n.d.). <https://www.bia.gov/bia/ots>

Pocket Neighborhoods • Creating small scale community in a large scale world. (n.d.). <https://www.pocket-neighborhoods.net/index.html>

Ruíz, M. A., & Mack-Vergara, Y. L. (2023). Resilient and Sustainable Housing Models against Climate Change: A Review. *Sustainability (Basel, Switzerland)*, 15(18), 13544-. <https://doi.org/10.3390/su151813544>

7 Directions Architects. (2023, November 22). *Stillaguamish village*. 7 Directions Architects |. <https://www.7directionsarchitects.com/project/stillaguamish-future-village/>

Q and A about Hud. HUD.gov / U.S. Department of Housing and Urban Development (HUD). (2021, March 25). <https://www.hud.gov/about/qaintro#>

Resources for tribal housing & tdhes. HUD.gov / U.S. Department of Housing and Urban Development (HUD). (n.d.-c).
https://www.hud.gov/program_offices/public_indian_housing/ih/codetalk/tribalhousing

Soak Up the Rain: Rain Gardens | US EPA. (2023, November 30). US EPA.
<https://www.epa.gov/soakuptherain/soak-rain-rain-gardens#:~:text=A%20rain%20garden%20is%20a,reduce%20runoff%20from%20your%20property.>

Title 15A: Lummi Nation Code of Laws, Flood Damage Protection Zone. (2005, July 22).

https://www.lummi-nsn.gov/userfiles/393_Title_15A_Flood_Damage_Prevention_Code_Resolution_2016-014.pdf

U.S. Census Bureau. (n.d.). *Explore Census data.*
https://data.census.gov/profile/Whatcom_County,_Washington?g=050XX00US53073#populations-and-people

U.S. Energy Information Administration - EIA - Independent Statistics and Analysis. (n.d.).
<https://www.eia.gov/state/analysis.php?sid=WA#53>

Wallace, C. L. (2017, April). *Architecture of the Salish Sea Tribes of the Pacific Northwest.* *fitch foundation.org*. Retrieved February 19, 2024, from https://fitchfoundation.org/wp-content/uploads/2017/05/FITCH_Christina-Wallace_final_web.pdf

